4.2 Building Trigonometry from an Integral

WORKSHEET

The question

In this question, you are not permitted to use any properties of trigonometric functions or inverse trigonometric functions.

The function T is defined for $x > 0$ by

$$\mathrm{T}(x) = \int_0^x \frac{1}{1+u^2}\,\mathrm{d}u,$$

and $\mathrm{T}_\infty = \displaystyle\int_0^\infty \frac{1}{1+u^2}\,\mathrm{d}u$ (which has a finite value).

(i) By making an appropriate substitution in the integral for $\mathrm{T}(x)$, show that

$$\mathrm{T}(x) = \mathrm{T}_\infty - \mathrm{T}(x^{-1}).$$

(ii) Let $v = \dfrac{u+a}{1-au}$, where a is a constant. Verify that, for $u \neq a^{-1}$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{1+v^2}{1+u^2}.$$

Hence show that, for $a > 0$ and $x < \dfrac{1}{a}$,

$$\mathrm{T}(x) = \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a).$$

Deduce that

$$\mathrm{T}(x^{-1}) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a^{-1})$$

and hence that, for $b > 0$ and $y > \dfrac{1}{b}$,

$$\mathrm{T}(y) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{y+b}{by-1}\right) - \mathrm{T}(b).$$

(iii) Use the above results to show that $\mathrm{T}(\sqrt{3}) = \frac{2}{3}\mathrm{T}_\infty$ and $\mathrm{T}(\sqrt{2}-1) = \frac{1}{4}\mathrm{T}_\infty$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Which trigonometric identities do parts (i) and (ii) correspond to? (Think of $\arctan x + \arctan \frac{1}{x}$, and the formula for $\tan(A+B)$.)

STAGE 2 The addition substitution

Ask yourself Differentiate $v = \dfrac{u+a}{1-au}$ by the quotient rule. Then compute $1+v^2$ as a single fraction and compare.

Ask yourself This says $\dfrac{\mathrm{d}u}{1+u^2} = \dfrac{\mathrm{d}v}{1+v^2}$. So the substitution $u \mapsto v$ doesn't change the integrand at all. What does it do to the limits of $\mathrm{T}(x)$?

STAGE 4 Special values

Pause and try Choose y and b in the last identity so that the argument $\frac{y+b}{by-1}$ equals y itself. What do you get?

Ask yourself For $\sqrt{2}-1$, try the addition identity with $x = a = \sqrt{2}-1$. What is $\frac{2x}{1-x^2}$? And what is $\mathrm{T}(1)$?

Try this next

1 **The exponential and the cosine.** (From the STEP II Specimen paper.) *Without using standard properties of exponential or trigonometric functions:* a function f satisfies $\mathrm{f}' = \mathrm{f}$ with $\mathrm{f}(0) = 1$. Show that $\frac{\mathrm{d}}{\mathrm{d}x}\big[\mathrm{f}(a-x)\mathrm{f}(x)\big] = 0$ and deduce that $\mathrm{f}(x)\mathrm{f}(y) = \mathrm{f}(x+y)$. Functions c and s satisfy $\mathrm{c}' = -\mathrm{s}$, $\mathrm{s}' = \mathrm{c}$, $\mathrm{s}(0) = 0$, $\mathrm{c}(0) = 1$. Show that $\mathrm{c}(x+y) = \mathrm{c}(x)\mathrm{c}(y) - \mathrm{s}(x)\mathrm{s}(y)$. (*Hint:* fix y and consider $\mathrm{c}(x+y) - \mathrm{c}(x)\mathrm{c}(y) + \mathrm{s}(x)\mathrm{s}(y)$ and a similar combination for s; show the sum of their squares is constant.)

2 **The inverse sine.** (From STEP III, 2007.) Let $\mathrm{S}(x) = \int_0^x \frac{\mathrm{d}u}{\sqrt{1-u^2}}$. Using $y = \frac{u}{\sqrt{1+u^2}}$, show that $\mathrm{T}(x) = \mathrm{S}\Big(\frac{x}{\sqrt{1+x^2}}\Big)$ and deduce that $\mathrm{S}\Big(\frac{1}{\sqrt{2}}\Big) = \frac{1}{2}\mathrm{T}_\infty$.

3 **An early Machin-type formula.** Use the addition identity to show that $\mathrm{T}(\frac{1}{2}) + \mathrm{T}(\frac{1}{3}) = \mathrm{T}(1) = \frac{1}{2}\mathrm{T}_\infty$.

4 T_∞ **is finite.** Show that $\mathrm{T}(x) \le 1 + \int_1^x \frac{\mathrm{d}u}{u^2} < 2$ for all $x \ge 1$, explain why $\mathrm{T}(x) < 2$ for $0 < x < 1$ as well, and deduce from the monotone convergence theorem (Section 2.2) that T_∞ exists.

Further reading

- **Tom Körner,** ***Calculus for the Ambitious*** (Cambridge University Press). Builds the elementary functions rigorously, discussing the different approaches.
- **Walter Rudin,** ***Principles of Mathematical Analysis*** (McGraw-Hill). The famous "baby Rudin"; the prologue to Chapter 8 constructs exp, sin and cos from power series in a few elegant pages. Advanced.
- **Victor Prasolov & Yuri Solovyev,** ***Elliptic Functions and Elliptic Integrals*** (American Mathematical Society). For where the lemniscate leads. Advanced.

The full guided solution, and what the question leads to at university, is in Section 4.2:
https://beyondstep.keithmcnulty.org/ch04/trig-from-integral.html